

Assignment 7: SQL Joins & Set Operations

Overview:

You will define two tables—Employees and Departments—and practice a variety of SQL joins and set operations to combine and compare their data.

1. Table Creation & Data Population

```
```sql
-- Employees table
CREATE TABLE Employees (
 EmpID INT PRIMARY KEY,
 EmpName VARCHAR(50),
 DepartmentID INT
);
INSERT INTO Employees (EmpID, EmpName, DepartmentID) VALUES
(1, 'Alice', 101),
(2, 'Bob', 102),
(3, 'Charlie', 103),
(4, 'Diana', NULL),
(5, 'Eve', 101);
```

**Submission Type :** Write the SQL query in the **PDF/Doc/Word** file and Take a screenshot of this part Output (**as screenshot\_1a**) and save it in the same **PDF/Doc/Word** file.

```
-- Departments table
CREATE TABLE Departments (
 DepartmentID INT PRIMARY KEY,
 DeptName VARCHAR(50),
 Location VARCHAR(50)
);
INSERT INTO Departments (DepartmentID, DeptName, Location) VALUES
(101, 'HR', 'New York'),
(102, 'IT', 'San Francisco'),
(103, 'Finance', 'Chicago'),
(104, 'Sales', 'Boston');
```

**Submission Type :** Write the SQL query in the **PDF/Doc/Word** file and Take a screenshot of this part Output (**as screenshot\_1b**) and save it in the same **PDF/Doc/Word** file.

## 2. SQL Tasks

1. **a) Inner Join:** List all employees with assigned department names.

SQL:

```
SELECT E.EmpID, E.EmpName, D.DeptName
FROM Employees E
INNER JOIN Departments D
ON E.DepartmentID = D.DepartmentID;
```

**Submission Type :** Write the SQL query in the **PDF/Doc/Word** file and Take a screenshot of this part Output (**as screenshot\_2a**) and save it in the same **PDF/Doc/Word** file.

2. **b) Left Join:** Show all employees, with department name if assigned (NULL if not).

SQL:

```
SELECT E.EmpID, E.EmpName, D.DeptName
FROM Employees E
LEFT JOIN Departments D
ON E.DepartmentID = D.DepartmentID;
```

**Submission Type :** Write the SQL query in the **PDF/Doc/Word** file and Take a screenshot of this part Output (**as screenshot\_2b**) and save it in the same **PDF/Doc/Word** file.

3. **c) Right Join:** Show all departments, with employee name if exists (NULL if none).

SQL:

```
SELECT D.DepartmentID, D.DeptName, E.EmpName
FROM Employees E
RIGHT JOIN Departments D
ON E.DepartmentID = D.DepartmentID;
```

**Submission Type :** Write the SQL query in the **PDF/Doc/Word** file and Take a screenshot of this part Output (**as screenshot\_2c**) and save it in the same **PDF/Doc/Word** file.

4. **d) Full Outer Join:** Include all employees and departments, matching where possible.

SQL:

```
SELECT E.EmpID, E.EmpName, D.DeptName
FROM Employees E
FULL OUTER JOIN Departments D
ON E.DepartmentID = D.DepartmentID;
```

**Submission Type :** Write the SQL query in the **PDF/Doc/Word** file and Take a screenshot of this part Output (**as screenshot\_2d**) and save it in the same **PDF/Doc/Word** file.

5. **e) Cross Join:** Generate every combination of employee names and department names.

SQL:

```
SELECT E.EmpName, D.DeptName
FROM Employees E
CROSS JOIN Departments D;
```

**Submission Type :** Write the SQL query in the **PDF/Doc/Word** file and Take a screenshot of this part Output (**as screenshot\_2e**) and save it in the same **PDF/Doc/Word** file.

6. **f) Union All:** Create a list combining all employee names and department names in one column.

SQL:

```
SELECT EmpName AS Name FROM Employees
UNION ALL
SELECT DeptName FROM Departments;
```

**Submission Type :** Write the SQL query in the **PDF/Doc/Word** file and Take a screenshot of this part Output (**as screenshot\_2f**) and save it in the same **PDF/Doc/Word** file.

7. **g) Intersect:** Find department IDs present in both tables.

SQL:

```
SELECT DepartmentID FROM Employees
INTERSECT
SELECT DepartmentID FROM Departments;
```

**Submission Type :** Write the SQL query in the **PDF/Doc/Word** file and Take a screenshot of this part Output (**as screenshot\_2g**) and save it in the same **PDF/Doc/Word** file.

8. **h) Except:** List department IDs in Departments but not in Employees.

SQL:

```
SELECT DepartmentID FROM Departments
EXCEPT
SELECT DepartmentID FROM Employees;
```

**Submission Type :** Write the SQL query in the **PDF/Doc/Word** file and Take a screenshot of this part Output (**as screenshot\_2h**) and save it in the same **PDF/Doc/Word** file.

Tip: If you encounter a "table already exists" error, append a suffix to the table name (e.g., Employees1).

### **Submission Guidelines:**

**Create One Main Folder (named [Assignment7\\_YourName](#)) which includes -**

- One **Doc/PDF/Word** file containing all the Queries and the screenshots of each part.
- Put your SQL file as **Assignment7\_YourName.sql**

**Convert the folder into ZIP format.**

**Upload the ZIP folder on the submission portal.**

**Watch this to learn how to create ZIP file**

[https://youtu.be/c7x-Q3AQEgw?si=\\_mS4c7OXADQ7DroE](https://youtu.be/c7x-Q3AQEgw?si=_mS4c7OXADQ7DroE)

**If you have any doubt, you can directly contact the Tutedude mentor.**